

Human Activity Recognition with Smart Devices

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Abstract—Human Activity Recognition (HAR) with mobile and wearable sensors plays an important role in fitness tracking and human-computer interaction. This study classifies activities such as walking, running, sitting, and stair climbing using accelerometer and gyroscope data from smartphones. Two approaches are applied: first feature extraction with traditional machine learning algorithms, and second, Recurrent Neural Networks (RNN) for direct time-series modeling. Experiments use public datasets (WISDM [1], KU-HAR [2]) with standardized preprocessing. Performance is evaluated through accuracy, precision, recall, F1-score, and confusion matrices, aiming for around %85 accuracy with traditional methods and above %95 with deep learning.

Link for the project: <https://github.com/aylinShn0/Human-Activity-Recognition-with-Smart-Devices>

WISDM dataset link:

<https://www.cis.fordham.edu/wisdm/dataset.php>

KU-HAR dataset link:

<https://www.kaggle.com/datasets/niloy333/kuhar/data>

Index Terms—Human Activity Recognition, Smart Devices, Machine Learning,

target is at least %85 accuracy for traditional methods and over %95 accuracy for deep learning models.

II. RELATED WORK

Human Activity Recognition (HAR) systems generally adopt two main machine learning approaches. The first is based on traditional, or shallow, algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Decision Trees, Naïve Bayes, and Perceptron, where feature extraction is performed manually [3], [4]. The second approach utilizes deep learning algorithms, including CNN, RNN, RBM, SAE, DFN where features are learned automatically within the hidden layers of the network

The processing pipeline typically begins with segmentation, in which the sensor data is divided into consecutive time windows, each analyzed independently. Segmentation can be implemented with overlapping or non-overlapping windows. Some studies employ a 50% overlap, while others use a non-overlapping strategy. The optimal configuration reported in the literature is a time window length of 2–5 seconds with a sampling frequency of 20–50 Hz [3].

For feature extraction, methods are often categorized into the time domain and the frequency domain. Time-domain features include statistical measures such as mean, minimum, maximum, standard deviation, median, root mean square, and signal magnitude area, while frequency-domain features are obtained via mathematical transformations that capture repetitive patterns [3]. Although both domains are used in HAR, the literature generally favors time-domain features due to their lower computational cost and better performance in many scenarios.

In dimensionality reduction, techniques such as PCA, LDA and KDA are widely explored. Among these, KDA has been reported to produce the best results for certain HAR datasets [3].

Regarding model training strategies, common methods include Cross-Validation, Leave-One-Subject-Out (LOSO), and a simple 70–30 train-test split. Cross-validation is the most frequently used approach due to its balanced bias-variance trade-off [3].

Performance metrics reported in the literature include Accuracy, Recall, Sensitivity, Specificity, Precision, and F1-score. General findings from prior work indicate that:

1) the best results are achieved when sensors are positioned at the waist,

2) the optimal sampling frequency is around 20 Hz,

I. INTRODUCTION

The widespread use of smartphones and wearable devices has enabled continuous monitoring of human physical activities, offering valuable insights for fitness and assisted living.

This project aims to develop an accurate and scalable HAR system using accelerometer and gyroscope data collected from mobile devices.

Accurate HAR systems can provide valuable insights for improving lifestyle, preventing health issues, and enabling personalized recommendations. In addition, automated activity recognition reduces the need for manual logging in fitness applications, enhancing user experience and engagement. The mix of mobile sensors and machine learning methods gives an effective and scalable solution. This project treats activity recognition as a classification task, where segments of sensor data are labeled with a specific, predefined activity.

The primary objective is to design, implement, and evaluate machine learning-based HAR models capable of achieving high classification accuracy. Two methodological approaches are explored:

1) Feature Extraction + Traditional ML Models – extracting statistical and signal-based features, followed by model training.

2) RNN-based Deep Learning – directly modeling time-series data without manual feature extraction.

The evaluation of the models will be based on accuracy, precision, recall, f1-score, confusion matrix. The performance

- 3) time-domain features often outperform frequency-domain features,
- 4) KDA provides superior dimensionality reduction performance, and
- 5) RNNs are more suitable for short-duration activities, while CNNs perform better for repetitive activities [3].

III. DATASET

This study utilizes two publicly available datasets for training and testing Human Activity Recognition (HAR) models: the WISDM dataset and the KU-HAR dataset. Both datasets are collected using smartphone sensors but differ significantly in data format, number of classes, sampling rates, and preprocessing requirements.

A. WISDM Dataset

The Wireless Sensor Data Mining (WISDM) dataset [1] contains raw accelerometer readings collected from smartphones at a sampling rate of 20 Hz. It comprises 1,098,207 labeled samples from 36 users, covering six activity classes: *Walking* (424,400 samples, 38.6%), *Jogging* (342,177, 31.2%), *Upstairs* (122,869, 11.2%), *Downstairs* (100,427, 9.1%), *Sitting* (59,939, 5.5%), and *Standing* (48,395, 4.4%). Each record follows the format:

```
[user], [activity], [timestamp],
[x-acceleration], [y-acceleration],
[z-acceleration]
```

The dataset contains no missing attribute values. It provides unprocessed sensor signals, making it ideal for manual segmentation and feature extraction in the time domain. For this study, WISDM was segmented into non-overlapping 3-second windows at 20 Hz, resulting in fixed-length segments of 60 samples per axis.

B. KU-HAR Dataset

The KU-HAR dataset [2] contains accelerometer and gyroscope readings from smartphones, collected from 90 participants performing 18 different activities. It comprises 1,945 raw activity samples (recorded in individual CSV files) and 20,750 fixed-length subsamples extracted from them. Activities include: *Stand*, *Sit*, *Talk-sit*, *Talk-stand*, *Stand-sit*, *Lay*, *Lay-stand*, *Pick*, *Jump*, *Push-up*, *Sit-up*, *Walk*, *Walk-backward*, *Walk-circle*, *Run*, *Stair-up*, *Stair-down*, and *Table-tennis*.

The dataset is organized into three main archives:

- 1) **Raw_time_domain_data.zip:** Original 1,945 time-domain samples
- 2) **Trimmed_interpolated_raw_data.zip:** Raw samples trimmed to remove extraneous segments and interpolated to a uniform sampling rate of 100 Hz.
- 3) **Time_domain_subsamples.zip:** 20,750 non-overlapping 3-second segments (100 Hz), stored in a single CSV file. Each segment includes:
 - Col. 1–300, 301–600, 601–900: accelerometer X, Y, Z
 - Col. 901–1200, 1201–1500, 1501–1800: gyroscope X, Y, Z

- Col. 1801: class ID (0–17)
- Col. 1802: length of each channel in the segment
- Col. 1803: subsample serial number

C. Comparison and Usage

While WISDM offers a large volume of raw, unprocessed accelerometer data at a lower sampling rate (20 Hz) and requires manual segmentation (3-second windows in this study), KU-HAR provides both raw and preprocessed multi-sensor data at a higher sampling rate (100 Hz) with a greater diversity of activities, already segmented into 3-second windows.

IV. METHODOLOGY

A. Method 1: Feature Extraction with Traditional Machine Learning

In the first approach, Human Activity Recognition (HAR) is treated as a supervised classification task by manually extracting statistical and signal-based features from segmented sensor data, followed by training multiple machine learning algorithms.

Data Splitting: Each dataset was split into training and testing sets using a 70–30 stratified split to preserve class distribution.

Segmentation: Sensor signals were segmented into fixed-length, non-overlapping 3-second windows. For the WISDM dataset, which has a 20 Hz sampling rate, each segment contained 60 samples per axis. For the KU-HAR dataset, pre-segmented data from the dataset was used. This dataset has a 100 Hz sampling rate, signals were segmented into fixed-length, non-overlapping 3-second windows.

Handling Missing Values: When missing sensor readings occurred they were imputed using linear interpolation based on the average of the immediately preceding and following values.

Feature Extraction: From each time window, the following features were computed for each axis: mean, minimum, maximum, standard deviation, median, median absolute deviation (MAD), root mean square (RMS), signal energy, and Shannon entropy. This produced a fixed-size feature vector per segment.

Feature Scaling: For models sensitive to feature magnitude (SVM, KNN, MLP), features were standardized using *StandardScaler* to zero mean and unit variance.

Dimensionality Reduction: PCA was optionally applied to retain 95% of the variance. This reduced feature dimensionality and allowed investigation of the trade-off between compression and performance. Models were trained in both PCA-reduced and original feature spaces.

Model Training: Six classifiers were evaluated: Support Vector Machine (SVM), Random Forest (RF), k-Nearest Neighbors (KNN), XGBoost, Multi-Layer Perceptron (MLP), and Naïve Bayes (NB). For each classifier, three configurations were tested:

- 1) *Without PCA:* Trained directly on extracted features.
- 2) *With PCA:* Trained on PCA-transformed features.

3) *With PCA + Hyperparameter Optimization:* PCA-transformed features followed by grid search over key hyperparameters.

Hyperparameter Optimization: Grid search was applied to tune model-specific parameters, including the number of neighbors in KNN, kernel type and regularization in SVM, number of estimators and depth in RF and XGBoost, and hidden layer sizes and activation functions in MLP. The goal was to maximize macro-averaged F1-score on the validation set.

Evaluation: Performance was evaluated using accuracy, macro-averaged F1-score, and confusion matrices.

B. Method 2: Recurrent Neural Network (RNN) for Direct Time-Series Modeling

The second approach models the HAR problem directly from raw segmented time-series data using Recurrent Neural Networks (RNNs), without manual feature extraction.

Segmentation: As in Method 1, sensor signals were divided into fixed-length, non-overlapping 3-second windows, with sampling frequency matched to the dataset.

Pure RNN Model: In the baseline RNN setup, the segmented multi-sensor time-series data was first processed by a SimpleRNN layer. This was followed by fully connected layers with dropout to prevent overfitting, and finally a softmax layer to predict the activity class. The model was trained using the categorical cross-entropy loss function and optimized with Adam.

Hybrid RNN + Feature Selection: To see if combining time-series modeling with handcrafted statistical features could improve results, the features from Method 1 (mean, min, max, standard deviation, median, MAD, RMS, energy, entropy) were calculated for each sensor axis and then added to the raw time-series data. A Random Forest classifier was used to choose the most important features. These selected features were scaled and repeated across the time dimension so their shape matched the RNN input, and then joined with the raw sensor data along the feature axis.

Hyperparameter Optimization: Key RNN settings — such as the number of recurrent units, dropout rate, learning rate, and batch size — were adjusted using some HAR deep learning studies [5], [6]. These studies also guided the choice of using a softmax output layer, the Adam optimizer, and categorical cross-entropy loss. The goal was to find a balance between model complexity and generalization, while preventing overfitting on the training data.

Evaluation: Both pure and hybrid RNN configurations were evaluated using accuracy, macro-averaged F1-score, and confusion matrices.

V. RESULTS

A. Method 1: Feature Extraction with Traditional Machine Learning

WISDM Dataset:

TABLE I
PERFORMANCE OF METHOD 1 ON WISDM DATASET

Model	PCA	Accuracy	Macro Avg F1-Score
SVM	No	0.90	0.87
Random Forest	No	0.93	0.91
KNN	No	0.93	0.91
XGBoost	No	0.94	0.93
MLP	No	0.93	0.91
Naïve Bayes	No	0.72	0.71
SVM	Yes	0.87	0.83
Random Forest	Yes	0.79	0.76
KNN	Yes	0.92	0.90
XGBoost	Yes	0.78	0.75
MLP	Yes	0.91	0.89
Naïve Bayes	Yes	0.68	0.57

For the WISDM dataset, classification experiments were performed on time-domain-based features extracted from 3-second non-overlapping time intervals. Three configurations were evaluated for each model: (i) without PCA, (ii) with PCA, and (iii) with PCA followed by hyperparameter optimization.

The highest performance was obtained by the XGBoost classifier without PCA, achieving an accuracy of 94% and a macro-averaged F1-score of 0.93. Following XGBoost, the

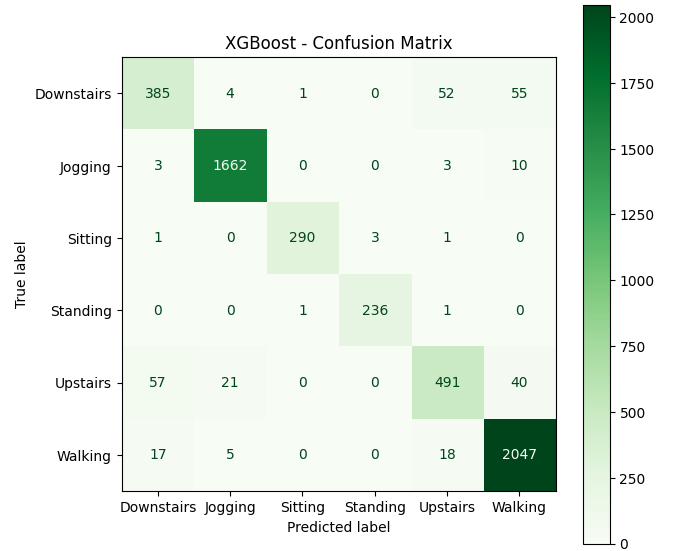


Fig. 1. Confusion Matrix for XGBoost without PCA on WISDM dataset.

best results were produced by the MLP (accuracy: 93%, F1=0.92) and KNN (accuracy: 93%, F1=0.91) classifiers, also without PCA. In contrast, the poorest performance was observed for Naïve Bayes in both PCA and non-PCA cases.

A detailed class-wise analysis of the top three models (XGBoost, MLP, and KNN) without PCA shows that all three achieve consistently high performance across most activity classes. For XGBoost, precision and recall values exceed 0.95 for all activities except *Downstairs* (precision: 0.83, recall: 0.77, F1-score: 0.80) and *Upstairs* (precision: 0.87, recall: 0.81, F1-score: 0.84). These two activities appear more challenging, likely due to their similarity in movement patterns and sensor signatures.

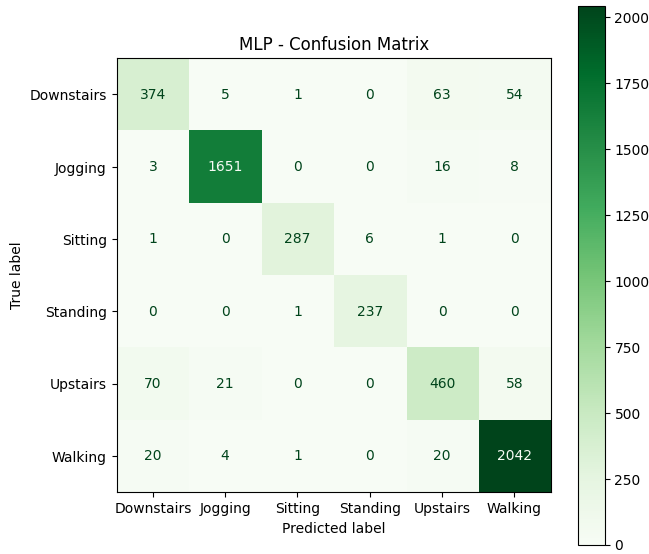


Fig. 2. Confusion Matrix for MLP without PCA on WISDM dataset.

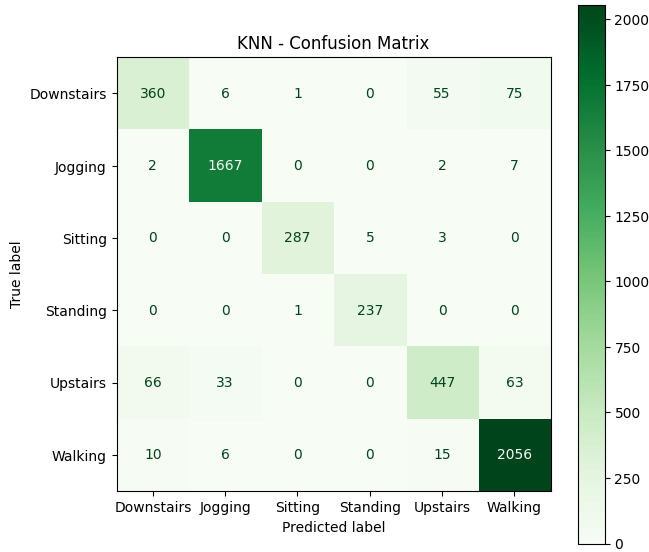


Fig. 3. Confusion Matrix for KNN without PCA on WISDM dataset.

MLP exhibits a similar trend, with very high scores for *Jogging*, *Sitting*, *Standing*, and *Walking*, but lower recall for *Downstairs* (0.75) and *Upstairs* (0.76), resulting in F1-scores of 0.78 and 0.79 for these classes, respectively.

KNN also achieves high precision and recall for most activities but shows the largest drop in recall for *Downstairs* (0.72) and *Upstairs* (0.73). This leads to slightly lower F1-scores of 0.77 and 0.79 for these classes compared to XGBoost and MLP.

Overall, while all three models perform exceptionally well for the majority of activities, confusion between stair-related activities (*Upstairs* and *Downstairs*) remains a common challenge. XGBoost maintains a slight edge in macro-averaged F1-score (0.93) over MLP (0.91) and KNN (0.91), confirming its

robustness in capturing both frequent and less frequent activity patterns.

TABLE II
CLASS-WISE PERFORMANCE OF XGBOOST ON WISDM DATASET (WITHOUT PCA)

Class	Prec.	Rec.	F1	Support
Downstairs	0.83	0.77	0.80	497
Jogging	0.98	0.99	0.99	1678
Sitting	0.99	0.98	0.99	295
Standing	0.99	0.99	0.99	238
Upstairs	0.87	0.81	0.84	609
Walking	0.95	0.98	0.97	2087
Accuracy	0.9458			
Macro Avg	0.94	0.92	0.93	5404
Weighted Avg	0.94	0.95	0.94	5404

TABLE III
CLASS-WISE PERFORMANCE OF MLP ON WISDM DATASET (WITHOUT PCA)

Class	Prec.	Rec.	F1	Support
Downstairs	0.80	0.75	0.78	497
Jogging	0.98	0.98	0.98	1678
Sitting	0.99	0.97	0.98	295
Standing	0.98	1.00	0.99	238
Upstairs	0.82	0.76	0.79	609
Walking	0.94	0.98	0.96	2087
Accuracy	0.9347			
Macro Avg	0.92	0.91	0.91	5404
Weighted Avg	0.93	0.93	0.93	5404

TABLE IV
CLASS-WISE PERFORMANCE OF KNN ON WISDM DATASET (WITHOUT PCA)

Class	Prec.	Rec.	F1	Support
Downstairs	0.82	0.72	0.77	497
Jogging	0.97	0.99	0.98	1678
Sitting	0.99	0.97	0.98	295
Standing	0.98	1.00	0.99	238
Upstairs	0.86	0.73	0.79	609
Walking	0.93	0.99	0.96	2087
Accuracy	0.9352			
Macro Avg	0.93	0.90	0.91	5404
Weighted Avg	0.93	0.94	0.93	5404

A detailed class-wise analysis of the top three models (XGBoost, MLP, and KNN) without PCA shows that all three achieve consistently high performance across most activity classes. For XGBoost, precision and recall values exceed 0.95 for all activities except Downstairs (precision: 0.83, recall: 0.77, F1-score: 0.80) and Upstairs (precision: 0.87, recall: 0.81, F1-score: 0.84). These two activities appear more challenging, likely due to their similarity in movement patterns and sensor signatures.

MLP exhibits a similar trend, with very high scores for Jogging, Sitting, Standing, and Walking but lower recall for Downstairs (0.75) and Upstairs (0.76), resulting in F1-scores of 0.78 and 0.79 for these classes, respectively.

KNN also achieves high precision and recall for most activities but shows the largest drop in recall for Downstairs

(0.72) and Upstairs (0.73). This leads to slightly lower F1-scores of 0.77 and 0.79 for these classes compared to XGBoost and MLP.

Overall, while all three models perform exceptionally well for the majority of activities, confusion between stair-related activities remains a common challenge. XGBoost maintains a slight edge in macro-averaged F1-score (0.93) over MLP (0.91) and KNN (0.91), confirming its robustness in capturing both frequent and less frequent activity patterns.

When PCA was applied, a general decrease in performance was observed for most models. The reduction was especially pronounced in tree-based algorithms such as Random Forest and XGBoost, where accuracy dropped to 77.81% and 77.54%, respectively. This suggests that PCA, caused information loss that harmed classification performance. KNN and MLP were less affected by PCA, showing only moderate performance degradation.

KU-HAR Dataset:

TABLE V
PERFORMANCE OF METHOD 1 ON KU-HAR DATASET

Model	PCA	Accuracy	Macro Avg F1-Score
SVM	No	0.3798	0.33
Random Forest	No	0.9443	0.94
KNN	No	0.3240	0.31
XGBoost	No	0.9582	0.95
MLP	No	0.6699	0.64
Naïve Bayes	No	0.1092	0.03
SVM	Yes	0.2752	0.20
Random Forest	Yes	0.2676	0.27
KNN	Yes	0.2182	0.20
XGBoost	Yes	0.2700	0.27
MLP	Yes	0.5198	0.47
Naïve Bayes	Yes	0.1092	0.03

The same experimental procedure was applied to the KU-HAR dataset. However, the results obtained for this dataset were generally lower compared to the WISDM dataset. This performance drop can largely be attributed to the higher complexity of KU-HAR, which contains a substantially larger number of activity classes (18 compared to WISDM's 6) and incorporates both accelerometer and gyroscope readings for each segment. The increased number of classes raises the classification difficulty and tends to reduce the performance of many traditional machine learning models.

Among all tested configurations, the best results were achieved by the XGBoost and Random Forest classifiers without PCA, achieving 95% and 94% accuracy, respectively, along with high macro-averaged F1-scores of 0.95 and 0.94. These results demonstrate that tree-based ensemble methods remain effective even in more challenging HAR scenarios with multi-sensor inputs, as they can model complex, non-linear decision boundaries and exploit heterogeneous feature distributions.

However, other models such as SVM, KNN, and MLP showed significantly lower scores. For example, SVM without PCA achieved only 37% accuracy (F1-score: 0.33), while KNN without PCA performed at 32% accuracy (F1-score:

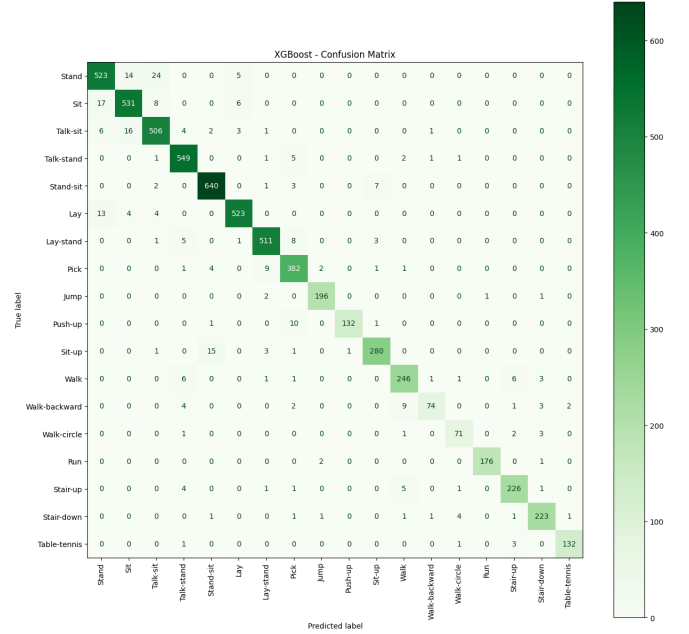


Fig. 4. Confusion Matrix for XGBoost without PCA on KU-HAR dataset.

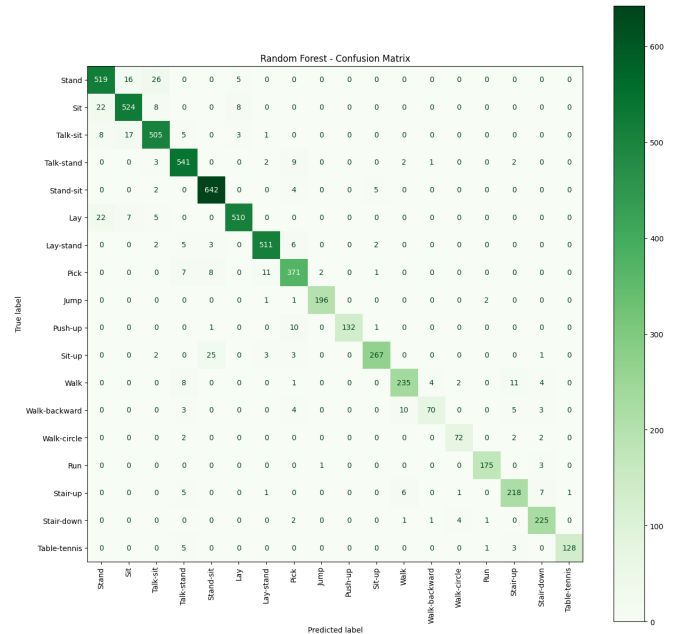


Fig. 5. Confusion Matrix for Random Forest without PCA on KU-HAR dataset.

0.31). MLP without PCA performed moderately better, reaching 66% accuracy (F1-score: 0.64), but still fell far short of the top-performing tree-based models.

The impact of PCA was again largely negative, with most models experiencing sharp performance drops. For example, applying PCA to Random Forest reduced accuracy from 94% to 26%, and for XGBoost, from 95% to 27%. These results confirm the trend observed in the WISDM dataset: PCA tends to harm performance in tree-based models when applied to HAR features, likely due to the removal of feature-specific variance crucial for decision splitting. As in the WISDM dataset, Naïve Bayes was unable to provide meaningful predictions, with accuracy around 10% in both PCA and non-PCA configurations, reflecting a near-random classification over 18 classes.

TABLE VI
CLASS-WISE PERFORMANCE OF XGBOOST (WITHOUT PCA) ON
KU-HAR DATASET

Class	Precision	Recall	F1-Score	Support
Stand	0.94	0.92	0.93	566
Sit	0.94	0.94	0.94	562
Talk-sit	0.93	0.94	0.93	539
Talk-stand	0.95	0.98	0.97	560
Stand-sit	0.97	0.98	0.97	653
Lay	0.97	0.96	0.97	544
Lay-stand	0.96	0.97	0.97	529
Pick	0.92	0.95	0.94	400
Jump	0.98	0.98	0.98	200
Push-up	0.99	0.92	0.95	144
Sit-up	0.96	0.93	0.94	301
Walk	0.93	0.93	0.93	265
Walk-backward	0.95	0.78	0.86	95
Walk-circle	0.90	0.91	0.90	78
Run	0.99	0.98	0.99	179
Stair-up	0.95	0.95	0.95	239
Stair-down	0.95	0.95	0.95	234
Table-tennis	0.98	0.96	0.97	137
Accuracy	0.9512			
Macro Avg	0.95	0.94	0.95	6225
Weighted Avg	0.95	0.95	0.95	6225

A detailed class-wise evaluation of Random Forest and XGBoost (both without PCA) on the KU-HAR dataset reveals that XGBoost outperforms Random Forest in both overall accuracy and macro-averaged F1-score. XGBoost achieved an accuracy of 95% and a macro F1-score of 0.95, while Random Forest reached 94% accuracy and 0.94 macro F1-score.

Both models consistently achieved high performance in most activity classes, with F1-scores above 0.90 for the majority of activities. However, certain class such as “Walk-backward” showed notably lower recall values (as low as 0.74 for Random Forest in “Walk-backward”). This suggests that these activities may share similar movement patterns with other classes, leading to misclassifications.

In high-performing classes like “Run,” “Stand-sit,” and “Talk-stand,” both models achieved near-perfect results, indicating their strong capability in recognizing distinct motion patterns. XGBoost demonstrated slightly more balanced precision-recall trade-offs across all classes, which reduced performance drops for harder activities.

TABLE VII
CLASS-WISE PERFORMANCE OF RANDOM FOREST (WITHOUT PCA) ON
KU-HAR DATASET

Class	Precision	Recall	F1-Score	Support
Stand	0.91	0.92	0.91	566
Sit	0.93	0.93	0.93	562
Talk-sit	0.91	0.94	0.92	539
Talk-stand	0.93	0.97	0.95	560
Stand-sit	0.95	0.98	0.96	653
Lay	0.97	0.94	0.95	544
Lay-stand	0.96	0.97	0.97	529
Pick	0.90	0.93	0.91	400
Jump	0.98	0.98	0.98	200
Push-up	1.00	0.92	0.96	144
Sit-up	0.97	0.89	0.93	301
Walk	0.93	0.89	0.91	265
Walk-backward	0.92	0.74	0.82	95
Walk-circle	0.91	0.92	0.92	78
Run	0.98	0.98	0.98	179
Stair-up	0.90	0.91	0.91	239
Stair-down	0.92	0.96	0.94	234
Table-tennis	0.99	0.93	0.96	137
Accuracy	0.9383			
Macro Avg	0.94	0.93	0.93	6225
Weighted Avg	0.94	0.94	0.94	6225

B. Method 2: Recurrent Neural Network (RNN) for Direct Time-Series Modeling

WISDM Dataset:

In the second approach, the WISDM dataset was evaluated using a SimpleRNN-based architecture. Hyperparameter search space was determined with guidance from two recent HAR studies [5], [6]. Initially, a grid search was considered, but due to the computational cost, random search was employed instead.

The hyperparameter search space was as follows:

- **units:** [30, 40, 60, 80]
- **dropout_rate:** [0.3, 0.5, 0.7]
- **batch_size:** [32, 64]

Fixed training parameters included 50 epochs, the Adam optimizer, and categorical cross-entropy loss. The number of random search iterations was set to 20.

The best-performing configuration was:

```
units = 30, dropout_rate = 0.3,
batch_size = 64
```

The best model achieved an overall accuracy of 79%, a macro-averaged F1-score of 0.65, and a weighted F1-score of 0.73. The confusion matrix (Fig. 6) reveals strong performance in Jogging and Walking classes, with recall values of 0.96 and 0.97, respectively. However, the model completely failed to correctly classify the Downstairs activity (precision = 0.00, recall = 0.00) and showed low recall for Upstairs (0.23). The Sitting and Standing classes exhibited balanced and high precision-recall values (precision $\geq$ 0.83, recall $\geq$ 0.90).

Overall, the SimpleRNN approach on WISDM demonstrated competitive results in certain activities such as *Jogging*, *Sitting*, and *Walking*, but struggled significantly with *Downstairs* and *Upstairs* classes. Compared to Method 1, the RNN-based approach achieved lower overall accuracy,

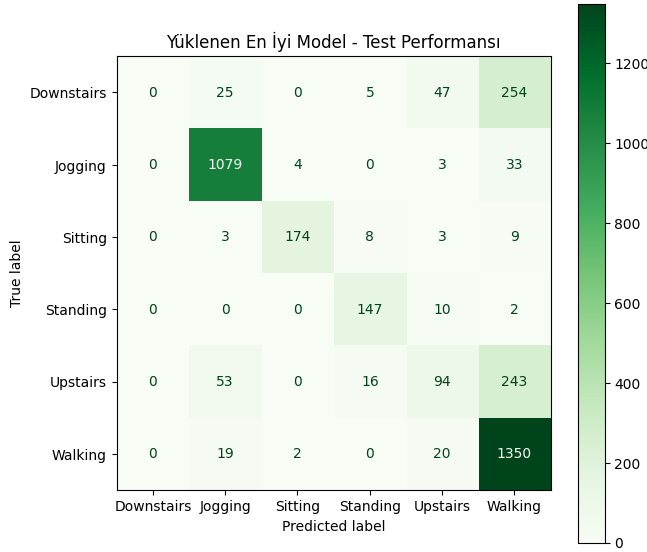


Fig. 6. Confusion Matrix for best SimpleRNN model on WISDM dataset.

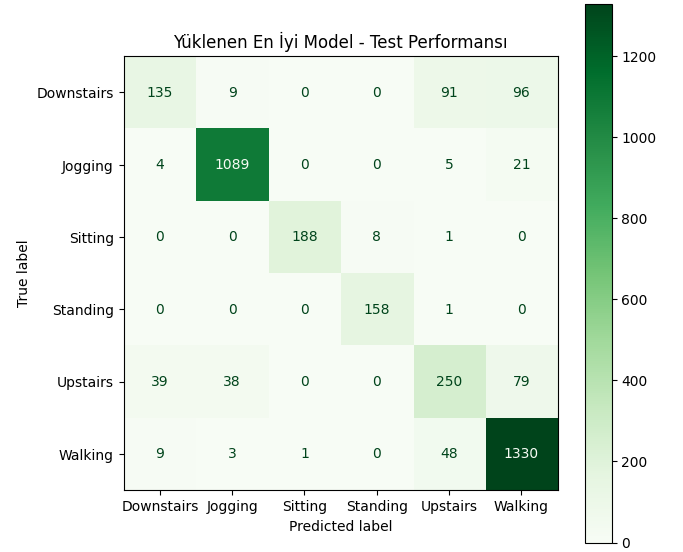


Fig. 7. Confusion Matrix for best Hybrid RNN model on WISDM dataset.

TABLE VIII
CLASS-WISE PERFORMANCE OF BEST SIMPLERNN MODEL ON WISDM DATASET

Class	Precision	Recall	F1-score	Support
Downstairs	0.00	0.00	0.00	334
Jogging	0.92	0.96	0.94	1119
Sitting	0.95	0.90	0.92	197
Standing	0.83	0.94	0.88	158
Upstairs	0.53	0.23	0.32	404
Walking	0.71	0.97	0.82	1391
Accuracy	0.79			
Macro Avg	0.66	0.67	0.65	3603
Weighted Avg	0.71	0.79	0.73	3603

TABLE IX
CLASS-WISE PERFORMANCE OF BEST HYBRID RNN MODEL ON WISDM DATASET

Class	Precision	Recall	F1-score	Support
Downstairs	0.76	0.35	0.48	334
Jogging	0.97	0.97	0.97	1119
Sitting	0.99	0.96	0.98	197
Standing	0.95	0.99	0.97	158
Upstairs	0.63	0.57	0.59	404
Walking	0.85	0.98	0.91	1391
Accuracy	0.87			
Macro Avg	0.86	0.80	0.82	3603
Weighted Avg	0.87	0.87	0.86	3603

indicating that handcrafted statistical features provided more discriminative information for this dataset.

After determining the best hyperparameters for the SimpleRNN model, these parameters were saved and reused in a hybrid configuration. In this configuration, raw segmented time-series data was augmented with the top 15 most important handcrafted features obtained via feature extraction and selection. The hybrid dataset was then used to train a SimpleRNN model with the previously determined optimal parameters.

Compared to the pure RNN model, the hybrid RNN with feature fusion achieved a substantial improvement in overall performance. The accuracy increased from 79% to 87%, and the macro-averaged F1-score rose from 0.65 to 0.82. The confusion matrix (Fig. 7) shows that this approach significantly improved recognition of the *Downstairs* class (recall from 0.00 to 0.35) and moderately enhanced the *Upstairs* class (recall from 0.23 to 0.57). High precision and recall were maintained for *Jogging*, *Sitting*, and *Standing*, all above 0.95 in both metrics.

These results indicate that the integration of handcrafted statistical features with raw time-series data can enhance the model's ability to discriminate between similar activity pat-

terns, leading to improved recognition of challenging classes such as *Downstairs* and *Upstairs*.

KU-HAR Dataset:

The same SimpleRNN-based workflow used for WISDM was applied to KU-HAR. The hyperparameter search used the identical ranges ($units \in \{30, 40, 60, 80\}$, $dropout_rate \in \{0.3, 0.5, 0.7\}$, $batch_size \in \{32, 64\}$), optimized with random search for 20 trials (50 epochs, Adam, categorical cross-entropy). The best configuration for KU-HAR was:

```
units = 30, dropout_rate = 0.3,
batch_size = 32
```

On KU-HAR, the SimpleRNN achieved an overall accuracy of 36%, a macro-averaged F1-score of 0.23, and a weighted F1-score of 0.29. The confusion matrix in Fig. 8 indicates that the model frequently confuses activity variants with similar kinematics (e.g., *Stand* vs. *Sit*, transitions like *Stand-sit*, and walking-related classes). Compared to WISDM (accuracy 79%, macro F1 = 0.65), performance is markedly lower. This degradation is expected given KU-HAR's higher class count (18 vs. 6), more heterogeneous motions, and multi-class transition activities that are harder to model with a shallow recurrent architecture.

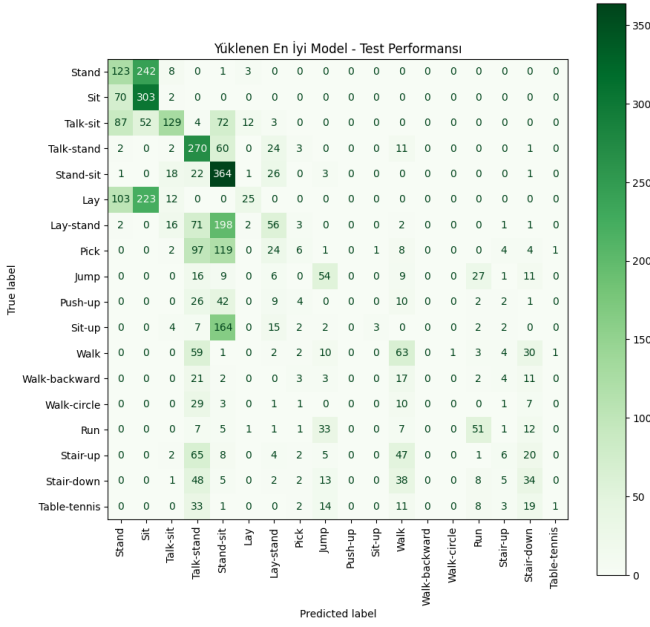


Fig. 8. Confusion Matrix for best SimpleRNN model on KU-HAR dataset.

TABLE X
CLASS-WISE PERFORMANCE OF BEST SIMPLERNN MODEL ON KU-HAR DATASET

Class	Precision	Recall	F1-score	Support
Stand	0.32	0.33	0.32	377
Sit	0.37	0.81	0.51	375
Talk-sit	0.66	0.36	0.46	359
Talk-stand	0.35	0.72	0.47	373
Stand-sit	0.35	0.83	0.49	436
Lay	0.57	0.07	0.12	363
Lay-stand	0.32	0.16	0.21	352
Pick	0.19	0.02	0.04	267
Jump	0.39	0.41	0.40	133
Push-up	0.00	0.00	0.00	96
Sit-up	0.75	0.01	0.03	201
Walk	0.27	0.36	0.31	176
Walk-backward	0.00	0.00	0.00	63
Walk-circle	0.00	0.00	0.00	52
Run	0.49	0.43	0.46	119
Stair-up	0.18	0.04	0.06	160
Stair-down	0.22	0.22	0.22	156
Table-tennis	0.33	0.01	0.02	92
Accuracy	0.36			
Macro Avg	0.32	0.27	0.23	4150
Weighted Avg	0.37	0.36	0.29	4150

While SimpleRNN yields reasonable results on WISDM, it underperforms on KU-HAR due to the larger class set and the presence of transitional/variant activities. This highlights the need for richer temporal modeling or feature fusion to better separate similar motion patterns on KU-HAR.

Following the same procedure as in WISDM, the best SimpleRNN parameters identified earlier were reused in a hybrid configuration. For KU-HAR, due to the dataset's six sensor channels (accelerometer and gyroscope XYZ), the top 24 most important handcrafted features (four per channel) were selected via feature importance ranking and concatenated

with the raw time-series signals along the feature axis.

Unlike in WISDM, where feature fusion improved classification performance, KU-HAR exhibited a noticeable decline in results. The hybrid RNN achieved an overall accuracy of only 28%, macro-averaged F1-score of 0.19, and weighted F1-score of 0.21, compared to the pure RNN baseline's 36% accuracy and 0.23 macro F1-score. The confusion matrix in Fig. 9 reveals that while the *Stand* class was classified almost perfectly (recall = 1.00), many other classes such as *Sit*, *Lay*, *Push-up*, *Walk-backward*, and *Walk-circle* suffered from very low or zero recall. This suggests that the feature fusion may have introduced noise or redundancy for this dataset's higher number of classes and complex inter-class similarities.

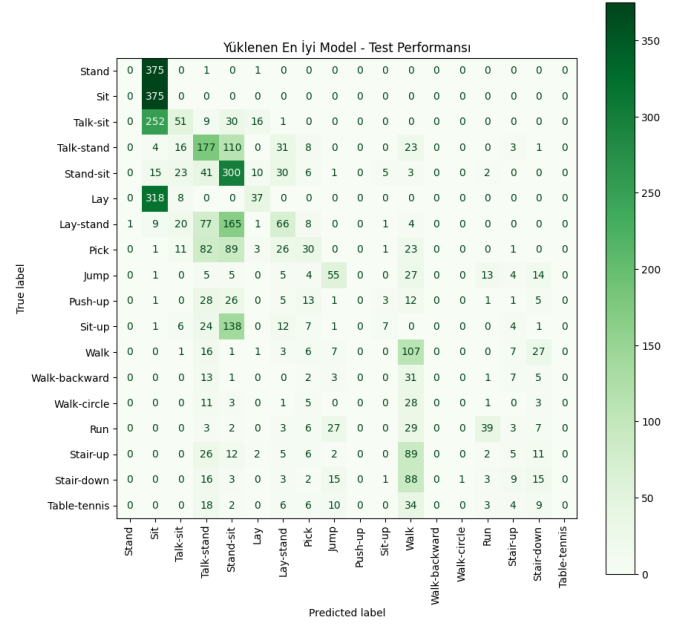


Fig. 9. Confusion Matrix for best Hybrid RNN model on KU-HAR dataset.

In WISDM, hybrid feature fusion boosted accuracy from 79% to 87% and macro F1 from 0.65 to 0.82, primarily by improving recognition of difficult classes like *Downstairs* and *Upstairs*. In KU-HAR, however, the same approach reduced performance, suggesting that the additional handcrafted features were less discriminative for its complex multi-class structure and may have hindered the temporal learning of the RNN.

VI. DISCUSSION

The experimental results highlight clear differences in the effectiveness of traditional machine learning approaches versus recurrent neural network-based time-series modeling for Human Activity Recognition (HAR).

For the WISDM dataset, Method 1 (feature extraction with traditional machine learning) consistently outperformed Method 2 (pure RNN) in terms of both accuracy and macro-averaged F1-score. XGBoost without PCA achieved the highest performance (94% accuracy, 0.93 macro-F1), followed closely by MLP and KNN. In contrast, the best SimpleRNN

TABLE XI
CLASS-WISE PERFORMANCE OF BEST HYBRID RNN MODEL ON
KU-HAR DATASET

Class	Precision	Recall	F1-score	Support
Stand	0.24	1.00	0.39	377
Sit	0.00	0.00	0.00	375
Talk-sit	0.32	0.07	0.11	359
Talk-stand	0.29	0.55	0.38	373
Stand-sit	0.33	0.58	0.42	436
Lay	0.15	0.01	0.03	363
Lay-stand	0.28	0.20	0.23	352
Pick	0.20	0.06	0.10	267
Jump	0.42	0.38	0.40	133
Push-up	0.00	0.00	0.00	96
Sit-up	0.48	0.06	0.11	201
Walk	0.26	0.25	0.26	176
Walk-backward	0.00	0.00	0.00	63
Walk-circle	0.00	0.00	0.00	52
Run	0.59	0.50	0.54	119
Stair-up	0.17	0.15	0.16	160
Stair-down	0.21	0.20	0.21	156
Table-tennis	0.20	0.01	0.02	92
Accuracy	0.28			
Macro Avg	0.23	0.22	0.19	4150
Weighted Avg	0.24	0.28	0.21	4150

configuration reached only 79% accuracy (0.65 macro-F1), struggling particularly with *Upstairs* and *Downstairs* classes due to their similar motion patterns. However, the hybrid RNN with handcrafted feature fusion improved classification of these difficult classes, increasing overall accuracy to 87% and macro-F1 to 0.82. This indicates that time-series models benefit from the additional discriminative power of statistical features when class boundaries are ambiguous.

For the KU-HAR dataset, the complexity of the task was significantly higher due to the increased number of activity classes (18 vs. 6 in WISDM), the inclusion of transitional activities, and multi-sensor data. Tree-based ensemble methods again dominated: XGBoost and Random Forest achieved 95% and 94% accuracy, respectively, without PCA. SVM, KNN, and MLP performed substantially worse, suggesting that linear and distance-based models struggle to capture the high intra-class variability of KU-HAR. The impact of PCA was consistently negative for tree-based models in both datasets, with accuracy drops exceeding 60% in KU-HAR, indicating that dimensionality reduction removed features critical for effective decision boundary construction.

The RNN-based approach performed poorly on KU-HAR (36% accuracy, 0.23 macro-F1), frequently misclassifying similar or transitional activities. Unlike in WISDM, the hybrid RNN with handcrafted features did not improve results, instead decreasing accuracy to 28%. This corruption may be caused by the addition of noisy or unnecessary features that disrupt the temporal learning process, particularly due to the large number of classes in KU-HAR and the complexity of interclass relationships.

Overall, the results suggest that:

- For datasets with fewer classes and clearer activity distinctions (WISDM), traditional machine learning with well-designed statistical features can match or outperform

simple deep learning models.

- Tree-based ensembles (XGBoost, Random Forest) are robust across both simple and complex HAR scenarios, likely due to their ability to model non-linear feature interactions.
- PCA is not beneficial for tree-based HAR models and can significantly harm performance by discarding informative variance.
- Feature selection with RNNs can improve performance for simpler datasets, but when applied to complex datasets with a high number of classes, it can hinder classification.

These findings highlight the importance of dataset characteristics in model selection and the need for tailored preprocessing and feature engineering strategies in HAR applications.

VII. CONCLUSION

In this study, Human Activity Recognition (HAR) was examined using accelerometer and gyroscope data from smartphones through two main approaches: (i) feature extraction with traditional machine learning algorithms, and (ii) direct time-series modeling with Recurrent Neural Networks (RNNs). Two publicly available datasets, WISDM [1] and KU-HAR [2], were used to test and compare these methods under different activity complexities and sensor setups.

Results showed that for datasets with fewer classes and simpler activity patterns, such as WISDM, traditional machine learning with handcrafted statistical features achieved strong results, with XGBoost reaching 94% accuracy and a macro-averaged F1-score of 0.93. For the more complex KU-HAR dataset, tree-based ensemble models such as XGBoost and Random Forest again delivered the best performance, achieving up to 95% accuracy. PCA-based dimensionality reduction consistently reduced performance in tree-based models, indicating that keeping the original feature variance is important for accurate classification.

The main contribution of this study is a detailed comparison of traditional feature-based methods and RNN-based models on datasets with different complexity levels, along with an analysis of how handcrafted features can support deep learning. The results also showed that combining handcrafted features with RNNs can improve recognition of challenging classes in simpler datasets, but may reduce performance in complex datasets with many classes.

Several aspects were not addressed in this work. Advanced deep learning architectures such as LSTMs, GRUs, CNN-LSTM hybrids, or attention mechanisms were not explored. Hyperparameter tuning was limited to a fixed search space, so potentially better configurations may have been missed. The study also did not investigate the effect of different sensor placements, data augmentation strategies, or cross-dataset testing, all of which could affect model robustness.

Future work may include testing more advanced temporal models, applying domain adaptation techniques for better cross-dataset performance. Real-time, on-device testing will also be important for evaluating the practical feasibility of HAR systems for mobile and wearable devices.

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